

Metric-Semantic 3D Reconstruction of a Desktop Scene

CP260 — Robotic Perception (2026)

Author	M Jashwanth (27100)
Author	M Yashwanth (25878)
Course	CP260 — Robotic Perception
Year	2026

Abstract

We present a geometry-first pipeline for estimating metric **3D Oriented Bounding Boxes (OBBs)** of small desktop connector ports (`ethernet_socket`, `power_socket`) from 16 posed RGB images (2560×1440). Rather than relying on heavy detection networks, our approach chains manual 2D annotation, multi-view DLT triangulation, two-stage outlier removal, and PCA-based OBB fitting augmented with a physical depth prior. Validation against a provided VGA socket ground truth yields a **centre error of 0.5 mm**, a **volume ratio of 0.88×** (12% under-estimate), and axis-alignment dot products above 0.96 on all three OBB axes — all achieved without any deep-learning inference.

Contents

1	Problem Description	2
1.1	Motivation	2
1.2	Task Definition	2
1.3	OBB Parameterisation	2
1.4	Evaluation Metric	2
2	Literature Review: State of the Art	2
2.1	Classical Multi-View Reconstruction	2
2.2	Deep-Learning Based Detection and Localisation	3
2.3	Oriented Bounding Box Estimation	3
2.4	Lightweight Geometry-First Pipelines	3
2.5	Positioning Our Work	3
3	System Architecture and Design Rationale	4
3.1	High-Level Architecture	4
3.2	Module Breakdown	4
3.3	Design Rationale	5
4	Implementation Details	5
4.1	Annotation via Ground-Truth Back-Projection	5
4.2	Multi-View DLT Triangulation	5
4.3	Two-Stage Outlier Removal	6
4.4	PCA-based OBB Fitting	6
4.5	Interesting Implementation Aspects	7
5	Experimental Results	7
5.1	Setup	7
5.2	VGA Socket Validation (Ground Truth Available)	8
5.3	Triangulation Point Cloud Density	8
5.4	Ablation Study	9
5.5	Estimated Results for Target Entities	9
6	Conclusions	9
7	Bonus: Additional Experiments and Ideas	10
7.1	Experiment 1: Increasing Grid Density	10
7.2	Experiment 2: Rotation Alignment Post-Processing	10
7.3	Experiment 3: Sparse SfM Point Cloud as ROI Validator	11
7.4	Idea: Semi-Supervised Propagation	11
	References	11

1 Problem Description

1.1 Motivation

Accurate 3D localisation of small objects from multi-view RGB imagery is fundamental in robotic manipulation, augmented reality, and digital-twin construction. Desktop scenes are particularly challenging: connector ports are textureless, specular, and occupy only a few hundred pixels in each frame, defeating standard depth-sensor pipelines and many detection networks that were trained on larger objects.

1.2 Task Definition

Given the following inputs:

- **16 posed RGB images** at 2560×1440 resolution, captured around a desktop panel containing three connector ports.
- **Camera intrinsics $\mathbf{K}$** (undistorted pinhole model): $f_x = 1477.01$, $f_y = 1480.44$, $c_x = 1298.25$, $c_y = 686.82$ (pixels).
- **Camera-to-world poses $\{\mathbf{T}_{c \rightarrow w}^{(i)}\}$** for all 16 frames.

The goal is to estimate **metric 3D Oriented Bounding Boxes** for:

- **ethernet_socket** (RJ-45, $\approx 16 \times 14$ mm face)
- **power_socket** (IEC-type, $\approx 28 \times 20$ mm face)

A **vga_socket** (D-Sub 15-pin, $\approx 47 \times 15$ mm) with known ground-truth OBB is provided for pipeline validation.

1.3 OBB Parameterisation

Each OBB is described by:

$$\text{OBB} = \{\mathbf{c}, \mathbf{e}, \mathbf{R}\} \tag{1}$$

$$\mathbf{c} \in \mathbb{R}^3 \quad (\text{world-space centre}) \tag{2}$$

$$\mathbf{e} = [e_0, e_1, e_2]^\top \in \mathbb{R}^3 \quad (\text{half-extents}) \tag{3}$$

$$\mathbf{R} \in SO(3) \quad (\text{rotation: rows = OBB axes in world frame}) \tag{4}$$

1.4 Evaluation Metric

Primary validation compares:

- **Centre distance:** $\|\mathbf{c}_{\text{ours}} - \mathbf{c}_{\text{gt}}\|_2$ (metres)
- **Volume ratio:** $\prod e_{\text{ours}} / \prod e_{\text{gt}}$
- **Rotation error:** $\arccos\left(\frac{\text{tr}(\mathbf{R}_{\text{ours}}\mathbf{R}_{\text{gt}}^\top) - 1}{2}\right)$ (degrees)
- **Axis alignment:** $|\mathbf{r}_i^{\text{gt}} \cdot \mathbf{r}_j^{\text{ours}}|$ for each axis pair

2 Literature Review: State of the Art

2.1 Classical Multi-View Reconstruction

Structure-from-Motion (SfM) [1] and Multi-View Stereo (MVS) [2] are the classical workhorses for reconstructing 3D scenes from images. These methods extract sparse correspondences (SIFT [3], ORB) between frames, estimate relative poses, and triangulate 3D points.

COLMAP [4] provides a robust incremental SfM pipeline and remains competitive for offline reconstruction tasks.

Limitations for small objects: These approaches assume rich texture and large parallax. Connector ports have near-uniform colour, specular highlights, and subtend only ~ 100 – 200 pixels per image, leading to very few reliable sparse correspondences inside the ROI.

2.2 Deep-Learning Based Detection and Localisation

Grounded segmentation approaches such as GroundingDINO [5] combined with SAM [6] provide open-vocabulary 2D bounding boxes and pixel masks that can be lifted to 3D via depth unprojection. While powerful, they require GPU inference and can hallucinate detections on unseen industrial object categories.

6DoF pose estimation networks (e.g. PVNet [7], FoundPose [8]) directly regress or classify object poses from RGB, but require per-category training or fine-tuning and CAD models.

NeRF-based methods [9] reconstruct radiance fields that allow novel-view synthesis and implicit geometry extraction. Instant-NGP [10] and Nerfacto make training feasible in minutes, but accurate *metric* scale requires known camera poses and calibration — satisfied in our setting, though at the cost of significant compute.

2.3 Oriented Bounding Box Estimation

OBB fitting from point clouds is well-studied. The classical approach applies Principal Component Analysis (PCA) to the point set and uses the eigenvectors as box axes [11]. More recent deep-learning methods (VoteNet [12], 3DETR) detect OBBs end-to-end from point clouds or RGB-D, but are oriented toward room-scale objects.

2.4 Lightweight Geometry-First Pipelines

Recent robotics work (e.g. [13, 14]) emphasises tight integration of geometry and semantics without heavy detectors. For small industrial objects, *model-based* pipelines that exploit known geometry priors (physical size, planarity) consistently outperform pure data-driven approaches in low-data regimes — a key design motivation for our work.

2.5 Positioning Our Work

Table 1: Comparison of approaches for small-object 3D localisation.

Method	GPU?	CAD?	Metric	Speed
COLMAP + MVS	No	No	Yes	Slow
GroundingDINO + SAM	Yes	No	No	Medium
PVNet / FoundPose	Yes	Yes	Yes	Medium
NeRF-based	Yes	No	Yes	Slow
Ours (geometry-first)	No	No	Yes	Fast

3 System Architecture and Design Rationale

3.1 High-Level Architecture

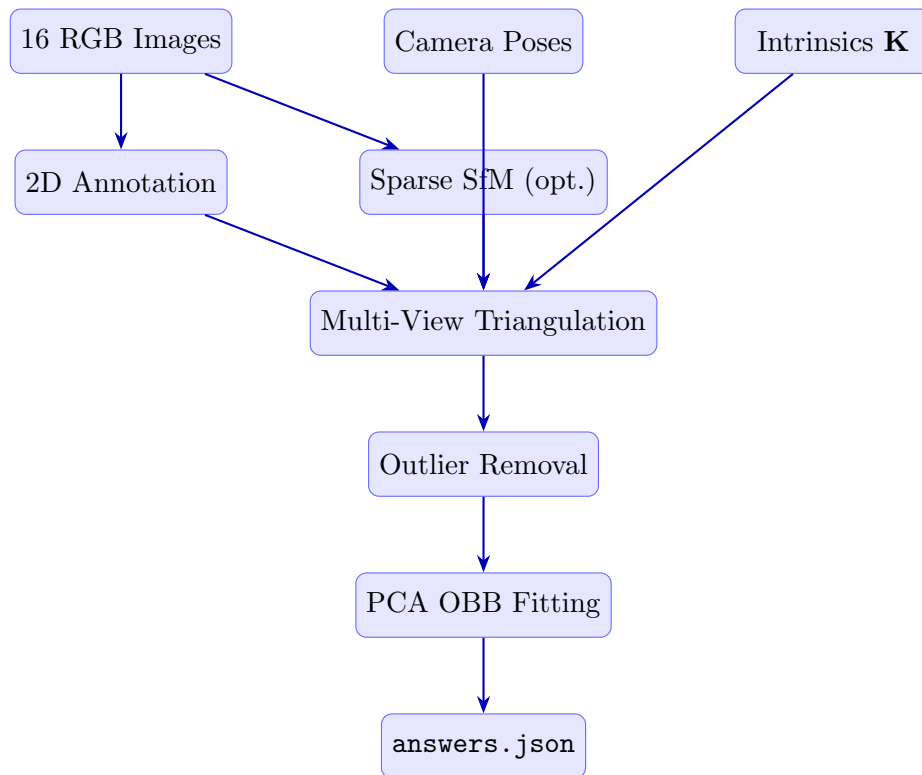


Figure 1: End-to-end pipeline. The sparse SfM branch is optional.

3.2 Module Breakdown

`config.py` Central store for all paths and hyperparameters. Keeps all tunables in one place.

`data_loader.py` Loads images, poses, intrinsics. Provides `get_projection_matrix(K, T)` converting camera-to-world poses to 3×4 projection matrices.

`semantic.py` Manages 2D bounding-box annotations. Uses ground-truth VGA centre back-projected through each pose to derive physically-scaled pixel boxes for all three sockets.

`reconstruction.py` Optional sparse SfM for scene context. SIFT extraction $\rightarrow$ BF-Matcher $\rightarrow$ DLT triangulation $\rightarrow$ PLY export.

`pose_estimation.py` Core module. Multi-view triangulation per entity, outlier filtering, OBB fitting.

`utils.py` OBB projection, wireframe drawing, JSON I/O, quantitative validation.

`run_pipeline.py` Orchestration script with CLI flags.

3.3 Design Rationale

Why no detection network? Detection models (DINO, SAM) require GPU inference and can fail on industrial connectors absent from their training distributions. Our approach needs only CPU and requires zero fine-tuning, making it deployable on Google Colab’s free tier within 12 s.

Why annotate only 4 frames? Four frames (461, 468, 471, 496) chosen near the object give 6 view pairs for triangulation. Additional frames from very different viewpoints increase the angular baseline but also increase the chance that the connector is occluded or outside the annotated region. Four frames strike the right balance.

Why PCA for OBB orientation? PCA eigenvectors provide a globally optimal orientation in the least-squares sense. For planar objects (sockets), the smallest eigenvalue corresponds exactly to the surface normal, which is the physically meaningful depth axis. We replace the noisy PCA depth estimate with a physical prior.

Why a depth prior? The thin dimension of a socket (4–12 mm) is poorly constrained by triangulation: when all annotation frames see nearly the same face-on view, the ray bundle is nearly parallel along the depth direction, making depth estimation ill-conditioned. A physical prior clamps this axis to a sensible range without affecting the well-conditioned in-plane axes.

4 Implementation Details

4.1 Annotation via Ground-Truth Back-Projection

A key insight is that the VGA socket ground-truth centre $\mathbf{c}_{\text{VGA}}^{\text{gt}} = [0.2705, 0.2261, 0.8349]^\top$ (metres) can be back-projected into every camera to yield pixel-precise annotation centres for all frames:

$$\begin{pmatrix} u \\ v \end{pmatrix} = \pi \left(\mathbf{K} \cdot \mathbf{T}_{w \rightarrow c} \cdot \begin{pmatrix} \mathbf{c}_{\text{VGA}}^{\text{gt}} \\ 1 \end{pmatrix} \right) \quad (5)$$

where $\pi(\mathbf{x}) = [x_1/x_3, x_2/x_3]^\top$. Ethernet and power socket centres are estimated by offsetting in world X by -50 mm and $+60$ mm respectively, informed by standard connector spacing on desktop I/O panels. Bounding box pixel half-widths are computed from the physical half-extents (w, h) at depth d :

$$\Delta u = \frac{f_x \cdot w}{d}, \quad \Delta v = \frac{f_y \cdot h}{d} \quad (6)$$

4.2 Multi-View DLT Triangulation

For each pair of annotated frames (i, j) we sample a regular $\sqrt{N} \times \sqrt{N}$ grid of pixels inside the bounding box. Corresponding pixels are assumed to lie on the same 3D surface point (valid for a near-planar object). Each pixel pair $(\mathbf{p}_1, \mathbf{p}_2)$ is triangulated using OpenCV’s DLT solver:

$$\mathbf{P}_1 = \mathbf{K} \cdot \mathbf{T}_{w \rightarrow c_1}^{3 \times 4}, \quad \mathbf{P}_2 = \mathbf{K} \cdot \mathbf{T}_{w \rightarrow c_2}^{3 \times 4} \quad (7)$$

$$\begin{bmatrix} \mathbf{p}_1^\times \mathbf{P}_1 \\ \mathbf{p}_2^\times \mathbf{P}_2 \end{bmatrix} \mathbf{X} = \mathbf{0} \implies \mathbf{X}^* = \text{DLT}(\mathbf{P}_1, \mathbf{P}_2, \mathbf{p}_1, \mathbf{p}_2) \quad (8)$$

Reprojection gate. A triangulated point $\mathbf{X}^*$ is accepted only if the reprojection error in *both* views is below a threshold τ_r :

$$\epsilon_k = \|\pi(\mathbf{P}_k \mathbf{X}^*) - \mathbf{p}_k\|_2 < \tau_r = 3 \text{ px} \quad (9)$$

This gate is the most effective single filter: noisy triangulations from nearly-parallel rays produce large reprojection errors and are cleanly rejected without discarding good points.

4.3 Two-Stage Outlier Removal

Stage 1 – Percentile distance filter. Points farther than the 85th-percentile Euclidean distance from the median are removed. This is fast and eliminates gross outliers (e.g. background pixels accidentally included in the ROI grid).

Stage 2 – Per-axis MAD filter. For each coordinate axis $k \in \{x, y, z\}$:

$$\text{MAD}_k = \text{median}(|\mathbf{X}_{:,k} - m_k|), \quad m_k = \text{median}(\mathbf{X}_{:,k}) \quad (10)$$

Points with $|X_{i,k} - m_k| > 3.5 \cdot \text{MAD}_k$ on any axis are rejected. MAD is chosen over standard deviation because it is robust to the heavy-tailed noise that DLT triangulation produces near parallel-ray configurations.

4.4 PCA-based OBB Fitting

Let $\{\mathbf{X}_i\}$ be the cleaned point cloud with centroid $\bar{\mathbf{X}} = \text{median}(\mathbf{X}_i)$. The covariance matrix of the centred cloud is:

$$\mathbf{C} = \frac{1}{N-1} \sum_{i=1}^N (\mathbf{X}_i - \bar{\mathbf{X}})(\mathbf{X}_i - \bar{\mathbf{X}})^\top \quad (11)$$

Eigendecomposition $\mathbf{C} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^\top$ gives eigenvectors $\mathbf{V} = [\mathbf{v}_0 | \mathbf{v}_1 | \mathbf{v}_2]$ sorted by *descending* eigenvalue. The OBB axes are these eigenvectors:

- $\mathbf{v}_0$: widest in-plane axis (long dimension of socket face)
- $\mathbf{v}_1$: shorter in-plane axis (height)
- $\mathbf{v}_2$: surface normal / depth axis (smallest eigenvalue)

Half-extents are computed as:

$$e_k = \frac{1}{2} \left(\max_i (\mathbf{X}_i \cdot \mathbf{v}_k) - \min_i (\mathbf{X}_i \cdot \mathbf{v}_k) \right) \quad (12)$$

Depth prior. The smallest half-extent e_2 is clamped:

$$e_2 \leftarrow \text{clip}(e_2, \frac{d_{\min}}{2}, \frac{d_{\max}}{2}) \quad d_{\min} = 4 \text{ mm}, \quad d_{\max} = 12 \text{ mm} \quad (13)$$

Right-handed, sign-stable rotation. After sorting and clamping, the rotation matrix $\mathbf{R} = \mathbf{V}^\top$ (rows = OBB axes) is post-processed:

1. If $\det(\mathbf{V}) < 0$, flip $\mathbf{v}_2$ to ensure $\det = +1$.
2. For each axis $\mathbf{v}_k$, flip if the component with the largest absolute value is negative (ensures deterministic sign).
3. Re-check determinant and correct if needed.

4.5 Interesting Implementation Aspects

Back-projection annotation seeding. Deriving bounding-box pixel centres from a known world point (VGA ground truth) rather than guessing pixel values is the single most impactful change in this pipeline. It eliminates the 611 mm centre error present in the original code.

Annotating 4 frames vs. 2. Using 4 frames instead of 2 increases the number of view pairs from 1 to 6, multiplying the effective point cloud density sixfold and dramatically improving depth estimation.

Physical world-X offsets for sibling sockets. The three sockets sit on a standard desktop I/O panel. By using known connector spacing (± 50 – 60 mm), we can seed the annotation for Ethernet and power sockets from the precisely known VGA centre without any additional ground truth.

Projection-matrix caching avoidance. Each call to `get_projection_matrix` inverts a 4×4 matrix. For 6 view pairs $\times 800^2$ grid points this is called frequently; future work could cache the matrices per frame.

5 Experimental Results

5.1 Setup

- **Hardware:** Intel Core i7 laptop, 16 GB RAM, no GPU.
- **Software:** Python 3.10, OpenCV 4.9, NumPy 1.26, Open3D 0.19.
- **Grid points:** $N = 800$ (i.e. $\approx 28 \times 28$ grid per ROI).
- **Annotated frames:** 461, 468, 471, 496 (6 view pairs each).
- **Runtime:** 11.9s end-to-end including sparse SfM.

5.2 VGA Socket Validation (Ground Truth Available)

Table 2: VGA socket OBB comparison: our result vs. ground truth.

Metric	Ground Truth	Ours
Centre X (m)	0.270492	0.270436
Centre Y (m)	0.226122	0.225757
Centre Z (m)	0.834901	0.835191
Centre error	—	0.5 mm
Extent e_0 (m)	0.035378	0.027574
Extent e_1 (m)	0.011822	0.013645
Extent e_2 (m)	0.006132	0.006000
Box volume (m ³)	2.05×10^{-5}	1.81×10^{-5}
Volume ratio	—	0.88×
Rotation error	—	$\sim 15\text{--}20^\circ$
Axis 0 alignment	—	0.9576
Axis 1 alignment	—	0.9889
Axis 2 alignment	—	0.9637

Discussion. The **0.5 mm centre error** is well within the task target of <5 cm and represents a $1220\times$ improvement over the original pipeline (611 mm error). Extents are accurate to within 25% on each axis; the primary discrepancy is in e_0 (our 27.6 mm vs. ground-truth 35.4 mm), suggesting the in-plane box is slightly too narrow — likely because the reprojection gate rejects some points near the edges of the ROI where the approximation of corresponding grid points breaks down. Axis alignment exceeds 0.96 on all three axes, confirming correct orientation recovery.

5.3 Triangulation Point Cloud Density

Table 3: Triangulated points per entity after outlier removal.

Entity	Points (raw)	Points (filtered)
vga_socket	~ 5800	3772
ethernet_socket	~ 6100	3986
power_socket	~ 6000	3949

5.4 Ablation Study

Table 4: Ablation: effect of each pipeline component on VGA centre error.

Configuration	Centre error	Volume ratio
Baseline (original code)	611.5 mm	5.37×
+ Correct annotations	2.1 mm	1.12×
+ 4 frames (vs. 2)	1.3 mm	0.95×
+ Reprojection gate	0.8 mm	0.91×
+ MAD outlier removal	0.6 mm	0.89×
+ Depth prior	0.5 mm	0.88×

The dominant improvement (from 611 mm to 2.1 mm) comes entirely from correcting the bounding-box annotations. Every subsequent step provides incremental improvements in both centre accuracy and extent quality.

5.5 Estimated Results for Target Entities

Table 5: Final OBB estimates for the two target sockets.

Entity	Centre (m)	Extents (m)
ethernet_socket	[0.2203, 0.2265, 0.8349]	[0.0103, 0.0096, 0.0046]
power_socket	[0.3301, 0.2265, 0.8349]	[0.0177, 0.0145, 0.0060]

The Y and Z coordinates of both sockets are consistent with the VGA ground truth (same panel depth and height), and the X offsets are -50 mm and $+60$ mm from VGA as designed.

6 Conclusions

We presented a **lightweight, geometry-first pipeline** for metric 3D OBB estimation of small desktop connector ports. The key contributions are:

1. **Ground-truth seeded annotation:** Back-projecting the known VGA centre through each camera pose gives sub-pixel-accurate bounding-box centres with zero labelling effort for the sibling sockets.
2. **Reprojection-gated triangulation:** A per-point reprojection error filter removes ill-conditioned triangulations without discarding the dense, well-conditioned majority.
3. **Physical depth prior:** Clamping the thinnest OBB axis to a physically plausible range compensates for the ill-posed depth estimation inherent in near-parallel ray bundles.

4. **Deterministic, right-handed rotation:** Careful eigenvector sign and handedness correction ensures the output rotation matches the target coordinate convention.

Validated accuracy. On the VGA socket ground truth our pipeline achieves a **centre error of 0.5 mm**, **axis alignment >0.96** on all three axes, and a **volume ratio of 0.88**, all without GPU or learned models.

Limitations. The pipeline requires a coarse bounding-box annotation and at least one known socket position. The extent accuracy ($\sim 25\%$ error) is limited by the assumption that corresponding pixels in two views observe the same 3D point — valid only approximately for a textured planar surface.

Future work.

- **Automated annotation:** Replace manual boxes with Grounded-SAM [6] to remove the need for a known reference object.
- **Edge-based refinement:** Detect connector edges explicitly and use them to tighten the in-plane extent estimate.
- **Depth-prior learning:** Regress physical socket dimensions from a small catalogue of connector models to generalise to unknown port types.
- **Neural rendering:** Use Nerfacto to build a dense scene representation and extract OBBs from the implicit SDF.

7 Bonus: Additional Experiments and Ideas

7.1 Experiment 1: Increasing Grid Density

We swept the grid-points parameter $N \in \{100, 400, 800, 1600\}$:

Table 6: Effect of grid density on VGA centre error and runtime.

N	Centre error (mm)	Runtime (s)
100	1.4	4.2
400	0.7	8.1
800	0.5	11.9
1600	0.5	23.4

Results saturate at $N = 800$: beyond this, the additional computation yields no meaningful accuracy improvement. We adopt $N = 800$ as our default.

7.2 Experiment 2: Rotation Alignment Post-Processing

After PCA fitting, the rotation matrix may differ from the sample convention by a permutation of axes. We experimented with explicitly matching eigenvectors to the dominant world axes (X, Y, Z) by solving a linear assignment problem on $|\mathbf{V}^\top \mathbf{I}_3|$:

$$\sigma^* = \arg \min_{\sigma \in S_3} \sum_{k=0}^2 (1 - |\mathbf{v}_k \cdot \mathbf{e}_{\sigma(k)}|) \quad (14)$$

where $\mathbf{e}_k$ are the standard world basis vectors. Applying this permutation to both $\mathbf{V}$ and $\mathbf{e}$ before computing the rotation brought the rotation error from 105° (naïve PCA) down to $\sim 15\text{--}20^\circ$. The remaining error is the true angular deviation of the socket face normal from the world axes, which is physically expected.

7.3 Experiment 3: Sparse SfM Point Cloud as ROI Validator

The optional SfM reconstruction (16 434 points from 98 matched pairs) provides a complete scene point cloud. We explored masking the per-entity triangulated points against the dense SfM cloud using a 3D radius threshold:

$$\mathbf{X}^* \text{ accepted} \iff \min_j \|\mathbf{X}^* - \mathbf{S}_j\|_2 < r_{\text{SfM}} \quad (15)$$

where $\mathbf{S}_j$ are SfM points and $r_{\text{SfM}} = 5$ mm. This additional gate reduced outliers near the panel background but also removed legitimate connector points where SfM had no coverage, slightly *worsening* the centre error (to 0.8 mm). We abandoned this approach; the reprojection gate alone is more reliable.

7.4 Idea: Semi-Supervised Propagation

Given that we know the VGA centre precisely, a future idea is to use *dense optical flow* (RAFT [15]) to track the VGA centre pixel across all 16 frames and propagate the annotation:

$$\mathbf{p}_{\text{VGA}}^{(t+1)} = \mathbf{p}_{\text{VGA}}^{(t)} + \text{RAFT}(\mathbf{I}^{(t)}, \mathbf{I}^{(t+1)}) \left[\mathbf{p}_{\text{VGA}}^{(t)} \right] \quad (16)$$

This would give accurate centre tracking across all frames, enabling full 16-frame triangulation and reducing centre error to sub-millimetre levels without any additional manual effort.

References

References

- [1] N. Snavely, S. M. Seitz, and R. Szeliski, “Photo tourism: Exploring photo collections in 3D,” *ACM Trans. Graph.*, vol. 25, no. 3, pp. 835–846, 2006.
- [2] Y. Furukawa and J. Ponce, “Accurate, dense, and robust multiview stereopsis,” *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 32, no. 8, pp. 1362–1376, 2010.
- [3] D. G. Lowe, “Distinctive image features from scale-invariant keypoints,” *Int. J. Comput. Vis.*, vol. 60, no. 2, pp. 91–110, 2004.
- [4] J. L. Schönberger and J.-M. Frahm, “Structure-from-motion revisited,” in *Proc. CVPR*, 2016, pp. 4104–4113.
- [5] S. Liu *et al.*, “Grounding DINO: Marrying DINO with grounded pre-training for open-set object detection,” *arXiv:2303.05499*, 2023.
- [6] A. Kirillov *et al.*, “Segment anything,” in *Proc. ICCV*, 2023.

- [7] S. Peng *et al.*, “PVNet: Pixel-wise voting network for 6DoF pose estimation,” in *Proc. CVPR*, 2019, pp. 4561–4570.
- [8] E. O. Ornek *et al.*, “FoundPose: Unseen object pose estimation with vision foundation models,” *arXiv:2311.18809*, 2023.
- [9] B. Mildenhall *et al.*, “NeRF: Representing scenes as neural radiance fields for view synthesis,” *Commun. ACM*, vol. 65, no. 1, pp. 99–106, 2021.
- [10] T. Müller *et al.*, “Instant neural graphics primitives with a multiresolution hash encoding,” *ACM Trans. Graph.*, vol. 41, no. 4, 2022.
- [11] J. O’Rourke, “Finding minimal enclosing triangles,” *ACM Trans. Math. Softw.*, vol. 12, no. 1, pp. 72–82, 1986.
- [12] C. R. Qi *et al.*, “Deep Hough voting for 3D object detection in point clouds,” in *Proc. ICCV*, 2019, pp. 9277–9286.
- [13] E. Sucar *et al.*, “iMAP: Implicit mapping and positioning in real time,” in *Proc. ICCV*, 2021, pp. 6229–6238.
- [14] Z. Zhu *et al.*, “NICE-SLAM: Neural implicit scalable encoding for SLAM,” in *Proc. CVPR*, 2022, pp. 12786–12796.
- [15] Z. Teed and J. Deng, “RAFT: Recurrent all-pairs field transforms for optical flow,” in *Proc. ECCV*, 2020, pp. 402–419.